TASK 1: Sprint Performance Metrics Analysis

Bepha Tools E-Commerce Project

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METRIC 1: DEFECT DENSITY

Formula: Defect Density = Total Defects ÷ Story Points Completed

Sprint 1: 3 defects ÷ 50 story points = **0.060 defects per story point**

Sprint 2: 5 defects ÷ 60 story points = **0.083 defects per story point**

Sprint 3: 2 defects ÷ 40 story points = **0.050 defects per story point**

Sprint	Total Defects	Story Points Completed	Defect Density (Defects ÷ SP)	Result
Sprint 1	3	50	3 ÷ 50	0.060 defects/SP
Sprint 2	5	60	5 ÷ 60	0.083 defects/SP
Sprint 3	2	40	2 ÷ 40	0.050 defects/SP

Insight: Sprint 2 had the highest defect density (0.083) due to AI feature complexity. Sprint 3 improved to the lowest density (0.050), showing better testing practices. Target for expansion Sprints: ≤ 0.050 defects/SP.

METRIC 2: AVERAGE CYCLE TIME

Formula: Average Cycle Time = Sum of all user story cycle times ÷ Number of user stories

Sprint 1: US1=3 days, US2=4 days, US3=3 days → (3+4+3) ÷ 3 = **3.33 days average**

Sprint 2: US1=5 days, US2=4 days, US3=2.5 days → (5+4+2.5) ÷ 3 = **3.83 days average**

Sprint 3: US1=2 days, US2=3 days, US3=2.5 days → (2+3+2.5) ÷ 3 = **2.50 days average**

Sprint	US1 Cycle Time	US2 Cycle Time	US3 Cycle Time	Calculation	Avg Cycle Time
Sprint 1	3 days	4 days	3 days	(3+4+3)÷3	3.33 days
Sprint 2	5 days	4 days	2.5 days	(5+4+2.5)÷3	3.83 days
Sprint 3	2 days	3 days	2.5 days	(2+3+2.5)÷3	2.50 days

Insight: Cycle time improved 35% from Sprint 2 (3.83 days) to Sprint 3 (2.50 days). The Sprint 2 spike was caused by unclear AI requirements and mid-sprint rework. Target for expansion Sprints: ≤ 2.50 days.

METRIC 3: AVERAGE LEAD TIME

Formula: Average Lead Time = Sum of all user story lead times ÷ Number of user stories

Sprint 1: US1=2 days, US2=3 days, US3=1.5 days → (2+3+1.5) ÷ 3 = **2.17 days average**

Sprint 2: US1=3 days, US2=4 days, US3=2.5 days → (3+4+2.5) ÷ 3 = **3.17 days average**

Sprint 3: US1=2 days, US2=3 days, US3=2.0 days → (2+3+2.0) ÷ 3 = **2.33 days average**

Sprint	US1 Lead Time	US2 Lead Time	US3 Lead Time	Calculation	Avg Lead Time
Sprint 1	2 days	3 days	1.5 days	(2+3+1.5)÷3	2.17 days
Sprint 2	3 days	4 days	2.5 days	(3+4+2.5)÷3	3.17 days
Sprint 3	2 days	3 days	2.0 days	(2+3+2.0)÷3	2.33 days

Insight: Lead time peaked in Sprint 2 (3.17 days) due to external blockers (payment gateway API delays, AI vendor docs). Sprint 3 recovered to 2.33 days. Target for expansion: ≤ 2.33 days.

METRIC 4: VELOCITY TRENDS

Sprint 1 Velocity: 50 story points completed

Sprint 2 Velocity: 60 story points completed (+20% vs Sprint 1)

Sprint 3 Velocity: 40 story points completed (–33% vs Sprint 2)

Average Velocity: (50 + 60 + 40) ÷ 3 = **50 story points per Sprint** ← Used for expansion planning

Sprint	Sprint Goal	SP Completed	vs Prior Sprint	Trend
Sprint 1	Foundation & Core Features	50 SP	(baseline)	Steady throughput
Sprint 2	AI Features & Order Tracking	60 SP	+10 SP (+20%)	Stretch sprint — complex AI features
Sprint 3	Customer Support & Reviews	40 SP	–20 SP (–33%)	Documentation & tech-debt focus
AVERAGE	All 3 Sprints	(50+60+40)÷3 = 50 SP	—	Planning baseline for expansion

Insight: Average velocity = 50 SP/Sprint. Sprint 2 peak reflects team capacity under ideal conditions. Sprint 3 dip reflects documentation overhead confirmed by team feedback. 50 SP/Sprint is the conservative, reliable planning baseline.

METRIC 5: WORK ITEM AGE TRENDS (Burndown Data)

Sprint 1 burndown: Day 1 = 50 remaining, Day 3 = 40, Day 5 = 30, Day 7 = 20, Day 10 = 0 (complete)

Sprint 2 burndown: Day 10 = 0 (complete) — all Sprint Goals fully achieved

Sprint 3 burndown: Day 10 = 0 (complete) — all Sprint Goals fully achieved

Day	Sprint 1 Stories Remaining	Sprint 2 Stories Remaining	Sprint 3 Stories Remaining	Work Item Age Assessment
Day 1	50	Not available in data	Not available in data	All stories newly created — age = 0
Day 3	40	—	—	Items aging 3+ days need monitoring
Day 5	30	—	—	Mid-sprint: any unstarted story is at risk
Day 7	20	—	—	Items remaining should be in active WIP
Day 10	0	0	0	All stories complete — no aged items at sprint end

Insight: Sprint 1 shows a consistent linear decline of ~10 stories every 2 days — a predictable, even pace. All three Sprints achieved 100% completion by Day 10 with zero aged items remaining. For expansion Sprints, flag any user story not started by Day 5 as an at-risk item requiring immediate Scrum Master attention.

QUALITATIVE DATA INSIGHTS

Sprint	Customer Satisfaction Findings	Scrum Team Satisfaction Findings
Sprint 1	Site intuitive; registration and catalog praised. Checkout could be faster.	Sprint goal clear; delivered on time. Testing felt rushed; needs better collaboration tools.
Sprint 2	AI assistant helpful but slow. Order tracking works. Custom tool design praised as innovative.	Proud of AI feature; workload heavy. Collaboration strong but tools insufficient. Planning inefficient.
Sprint 3	Customer support responsive; reviews helpful. Search accuracy needs improvement.	Sprint felt smooth; workload balanced. Communication improved. Need more time for docs and tech debt.